

Classification of Bachelor Thesis – opponent

Author of classification: Ing. Petr Lukáš, Ph.D.
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Opponents: Ing. Petr Lukáš, Ph.D.
Title: Trie Data Structure
Thesis version: 1
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1. *Meeting the requirements of the thesis assignment.*

The difficulty of the work corresponds to a typical bachelor thesis and the work fully meets the requirements.

2. *Thesis technicality evaluation.*

The structure of the chapters is logical and the individual parts follow each other appropriately. From a linguistic perspective, the work contains several grammatical mistakes (e.g., inappropriate verb forms or prepositions, missing words, etc.) For example, in the sentence "... a node POINTS to child nodes is the parent node" (page 12), POINTING should be used instead of POINTS. Nevertheless, the text is understandable and easy to read.

3. *Results evaluation of the thesis.*

The student implemented several variants of the trie data structures, including a simple test. I did not notice any errors or malfunctions during testing; however, I have a few complaints about the quality of the source code. Individual variants of the trie are implemented as separate classes, which all share methods with a similar signature. I am missing some abstract class that would guarantee a common interface for these classes. All source codes are in .cpp files without using header files. There is a memory leak - individual nodes of the trie data structures are not deallocated. Several constants are hard-coded in the source code (e.g. the load factor of the hash table). From the practical point of view, it would be better, if the methods for the prefix searching returned an iterator instead of vector of strings. The iterator would avoid unnecessary materialization of the result.

4. *Evaluation of the new findings contribution.*

The work does not introduce new findings. The implemented classes would be utilizable in practice after minor fixes (especially the memory leak).

5. *Utilization and selection of information sources.*

I have no complaints about the selection of the information sources. All quoted parts are clearly distinguished from the rest of the text.

6. *Question for the defense of the thesis.*

Would it be possible to avoid the result materialization in the prefix searching using an iterator?

7. *Summary evaluation.*

The overall result of the work is, in my opinion, very good.

Overall classification: very good

Město Albrechtice, 22.05.2025

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